

KAWAGUCHIKO, Japan

WMO: 476400

Lat: 35.5006N

Lon: 138.7608E

Elev: 2826

StdP: 13.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	19.3	-0.9	5.8	28.3	1.7	6.6	29.1	16.2	31.6	14.2	32.1	3.0	240	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	86.5	72.2	84.2	71.5	81.7	70.5	74.0	82.8	72.9	80.9	71.8	79.1	6.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	129.6	78.2	70.4	124.6	77.2	69.4	120.1	76.1	39.7	83.1	38.6	81.2	37.5	79.3	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.5	9.9	DB	10.5	91.1	2.7	1.7	8.6	92.3	7.0	93.3	5.5	94.2	3.5	95.4	(4)
(5)			WB	9.0	76.3	2.6	1.4	7.1	77.3	5.6	78.2	4.2	79.0	2.3	80.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	31.6	33.4	40.0	49.5	58.5	64.7	72.5	73.8	66.6	56.4	46.2	36.9	(6)	
	DBStd	15.44	4.85	5.96	6.29	6.67	5.00	4.88	4.51	3.92	5.51	5.61	5.84	5.48	(7)	
	HDD50	2006	571	466	317	88	3	0	0	0	0	8	145	408	(8)	
	HDD65	5197	1036	885	775	465	209	63	4	3	48	273	564	872	(9)	
	CDD50	2963	0	1	7	73	267	440	698	739	499	207	30	2	(10)	
	CDD65	679	0	0	0	0	8	54	237	277	96	7	0	0	(11)	
	CDH74	3632	0	0	0	9	76	240	1338	1556	387	26	0	0	(12)	
	CDH80	894	0	0	0	1	5	34	376	418	58	2	0	0	(13)	
(14) Wind	WSAvg	4.2	4.3	4.5	4.8	4.9	4.5	4.0	4.1	4.0	3.9	3.7	3.7	4.2	(14)	
(15) Precipitation	PrecAvg	56.9	2.1	2.1	3.6	3.9	4.7	6.0	6.3	6.9	9.2	6.9	3.2	1.9	(15)	
	PrecMax	83.0	6.0	4.6	6.7	9.1	10.9	10.6	12.5	16.3	20.1	26.0	9.7	5.1	(16)	
	PrecMin	34.1	0.1	0.1	1.9	1.2	2.0	2.7	2.1	1.4	2.1	0.8	0.1	0.2	(17)	
	PrecStd	10.3	1.6	1.1	1.3	1.9	2.1	2.2	3.0	3.9	5.0	5.1	2.1	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	58.5	65.7	74.6	80.0	83.9	90.0	89.4	84.7	77.3	67.4	59.5	(19)	
		MCWB	42.8	48.3	50.5	56.6	62.3	68.9	72.0	72.9	71.2	64.0	55.5	49.5	(20)	
	2%	DB	48.2	52.8	60.2	69.3	76.0	79.7	86.8	86.8	81.5	72.4	63.6	54.3	(21)	
		MCWB	40.2	44.5	47.6	54.2	60.4	67.1	72.0	72.7	69.5	62.8	53.2	45.9	(22)	
	5%	DB	44.6	48.3	56.0	65.4	72.5	76.6	84.3	84.6	78.4	69.0	60.0	50.9	(23)	
		MCWB	36.4	40.1	45.9	52.4	58.5	65.4	71.4	72.1	68.2	60.3	51.1	43.0	(24)	
	10%	DB	41.5	44.5	52.3	62.1	69.4	73.6	81.5	82.2	75.5	66.1	56.9	47.6	(25)	
		MCWB	33.7	37.1	44.0	51.0	57.7	64.2	70.8	71.5	67.0	58.9	49.5	40.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	51.0	55.1	61.9	66.9	71.9	75.4	75.7	73.0	67.4	59.3	53.7	(27)	
		MCDB	50.1	55.2	60.3	66.9	74.1	80.1	85.6	85.2	81.2	73.6	63.5	56.3	(28)	
	2%	WB	41.4	46.1	51.0	58.2	63.8	69.5	73.8	74.4	71.0	64.8	56.2	48.3	(29)	
		MCDB	45.9	52.3	56.1	65.4	70.9	76.6	82.8	83.3	78.3	70.2	60.3	51.9	(30)	
	5%	WB	37.3	41.4	48.2	55.4	61.7	67.8	72.7	73.3	69.7	62.5	53.5	44.4	(31)	
		MCDB	43.0	46.3	53.8	61.7	68.7	73.6	81.0	81.3	75.8	67.1	57.4	48.8	(32)	
	10%	WB	34.5	37.9	44.9	53.0	59.5	66.2	71.5	72.3	68.4	60.1	51.1	41.2	(33)	
		MCDB	40.3	42.9	51.1	59.6	66.7	71.5	79.0	79.6	73.9	64.6	55.5	46.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	18.8	19.4	20.4	19.2	15.2	15.0	15.6	14.7	15.9	19.0	18.8	(35)	
		MCDBR	22.3	24.0	25.5	25.5	24.7	20.1	19.7	19.2	18.8	19.5	22.1	22.7	(36)	
	5% WB	MCWBR	15.0	16.4	16.1	13.8	12.4	9.3	8.0	7.9	8.8	10.7	13.4	15.1	(37)	
		MCWBR	21.1	21.4	22.5	21.1	20.3	16.0	17.9	17.6	16.2	16.6	18.9	19.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.330	0.374	0.430	0.481	0.506	0.539	0.528	0.529	0.467	0.430	0.368	0.328	(40)	
		taud	2.335	2.201	2.054	1.961	1.948	1.928	2.025	2.011	2.150	2.177	2.302	2.378	(41)	
	Ebn at Noon		272	271	264	257	252	243	245	241	251	248	256	266	(42)	
		Edh at Noon	31	39	50	58	59	60	55	54	45	40	32	28	(43)	
(44) All-Sky Solar Radiation	RadAvg		906	1090	1335	1531	1635	1425	1501	1512	1217	1007	846	792	(44)	
	RadStd		50	86	86	153	167	121	223	134	110	106	73	58	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page